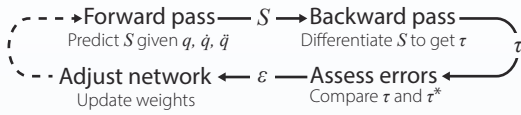


Inverse Dynamics

1. Train the network

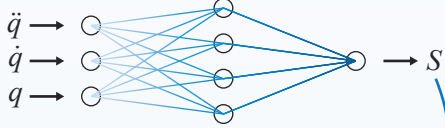
Approximate Appell's function S by minimizing the error between its derivatives and reference torques τ^* :



simulation

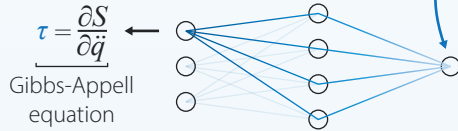
2. Forward pass

Evaluate the trained network to map the state and acceleration inputs $(q, \dot{q}, \ddot{q})$ to the scalar S :



3. Backward pass

Compute torques τ by extracting the gradient of S with respect to acceleration:

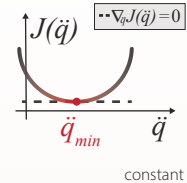


Forward Dynamics

1. Define objective

Minimum of cost function $J(\ddot{q})$ satisfies Gibbs-Appell equation:

$$\nabla_{\ddot{q}} J(\ddot{q}) = \frac{\partial S}{\partial \ddot{q}} - \tau^* = 0$$



$$J(\ddot{q}) = \int \left(\frac{\partial S}{\partial \ddot{q}} - \tau^* \right) d\ddot{q} = S(q, \dot{q}, \ddot{q}) - \tau^* \ddot{q} + \dots$$

simulation

2. Solve for acceleration

Find the acceleration $\ddot{q}_{min}$ that minimizes the cost $J(\ddot{q})$:

$$\ddot{q}_{min} = \underset{\ddot{q}}{\operatorname{argmin}} \left[\overbrace{S(q, \dot{q}, \ddot{q})}^{\text{initial conditions}} - \underbrace{\tau^* \ddot{q}}_{\substack{\text{network from} \\ \text{inverse dynamics step}}} + \underbrace{\dots}_{\text{regularization term}} \right]$$

3. Step forward

Integrate found acceleration $\ddot{q}_{min}$ to get the next state:

$$q_{t+\Delta t}, \dot{q}_{t+\Delta t} \leftarrow \int \ddot{q}_{min} dt$$